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Jnana Sangama, Belagavi - 590018



A Project Work Phase-I (BCS685)

Report on

“INDIAN MARKET PORTFOLIO INTELLIGENCE AND BACKTESTING SYSYTEM”

*Project Report submitted in partial fulfilment of the requirement for the
award of the degree of*

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

ARYAMAN TIWARI	1KS23CS020
CHIRAG T	1KS23CS034
DURGASHREE M	1KS23CS043
JAYADITYA DEV	1KS23CS059

Under the guidance of

Mrs. Beena K

Assistant Professor

Department of Computer Science & Engineering

K.S.I.T, Bengaluru-560109



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

K. S. Institute of Technology

An Autonomous Institution Under VTU, Belagavi

#14, Raghuvanahalli, Kanakapura Road, Bengaluru - 560109

2025 - 2026

K. S. Institute of Technology

An Autonomous Institution Under VTU, Belagavi
#14, Raghuvanahalli, Kanakapura Road, Bengaluru - 560109

Department of Computer Science & Engineering



CERTIFICATE

Certified that the Project Work Phase-I (BCS685) entitled “**INDIAN MARKET PORTFOLIO INTELLIGENCE AND BACKTESTING SYSYTEM**” is bonafide work carried out by:

ARYAMAN TIWARI	1KS23CS020
CHIRAG T	1KS23CS034
DURGASHREE M	1KS23CS043
JAYADITYA DEV	1KS23CS059

in partial fulfillment for VI semester B.E., Project Work in the branch of Computer Science and Engineering prescribed by **Visvesvaraya Technological University, Belagavi** during the period of January 2026 to May 2026. It is certified that all the corrections and suggestions indicated for internal assessment have been incorporated. The Project Work Phase-I Report has been approved as it satisfies the academic requirements in report of project work prescribed for the Bachelor of Engineering degree.

.....
Signature of the Guide

[Mrs. Beena K]

.....
Signature of the HOD

[Dr. Rekha B Venkatapur]

.....
**Signature of the Principal
& Director**

[Dr. Dilip Kumar K]

DECLARATION

We, the undersigned students of 6th semester, Computer Science & Engineering, KSIT, declare that our Project Work Phase-I entitled “**INDIAN MARKET PORTFOLIO INTELLIGENCE AND BACKTESTING SYSYTEM**”, is a bonafide work of ours. Our project is neither a copy nor by means a modification of any other engineering project.

We also declare that this project was not entitled for submission to any other university in the past and shall remain the only submission made and will not be submitted by us to any other university in the future.

Place:

Date:

Name and USN

Signature

Aryaman Tiwari (1KS23CS059)

Chirag T (1KS23CS034)

Durgashree M (1KS23CS043)

Jayaditya Dev (1KS23CS059)

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**ARYAMAN TIWARI
CHIRAG T
DURGASHREE M
JAYADITYA DEV**

ABSTRACT

The rapid growth of financial markets and the increasing participation of retail investors have created a strong demand for intelligent tools that can assist in investment analysis and decision-making. Traditional investment approaches often rely on intuition or limited market analysis, which may lead to higher financial risks and poor strategy selection. To address these challenges, this project presents an Indian Market Portfolio Intelligence and Back testing Platform, a machine learning-based system designed to evaluate trading strategies using historical stock market data.

The proposed platform integrates portfolio analysis, strategy back testing, market regime detection, and risk assessment into a unified framework. Historical market data is collected dynamically using the Yahoo Finance API and processed to generate important financial indicators such as returns, volatility, momentum, and maximum drawdown. The system supports multiple predefined trading strategies including Buy and Hold, Moving Average Crossover, RSI-based trading, Momentum Strategy, and Mean Reversion Strategy. These strategies are tested against historical market conditions to evaluate their effectiveness and performance.

Machine learning techniques are incorporated to improve strategy analysis and recommendation. Unsupervised learning methods are used for market regime detection, enabling the identification of bullish, bearish, and sideways market conditions. Supervised learning models are employed to predict the suitability of trading strategies under varying market scenarios. In addition, the platform performs risk and performance analysis by calculating important financial metrics such as Sharpe Ratio, volatility, drawdown, and cumulative returns.

The system is designed to benefit investors, researchers, and students by providing data-driven insights into stock market behaviour and trading performance. By combining financial analytics with machine learning, the platform enhances investment decision support and promotes a better understanding of portfolio management and risk evaluation in the Indian stock market environment.

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Chapter 1

INTRODUCTION

The stock market plays a major role in the global economy by enabling individuals and organizations to invest, trade, and grow their financial assets. In recent years, the Indian stock market has experienced rapid growth due to increased participation from retail investors, digital trading platforms, and the availability of real-time financial data. However, financial markets are highly dynamic and unpredictable, making investment decisions difficult for both beginners and experienced investors. Selecting an appropriate trading strategy without proper

Traditionally, investors rely on technical indicators, historical price charts, or market intuition to make trading decisions. While these methods provide useful insights, they often fail to evaluate the robustness of a strategy under different market conditions. Existing platforms mainly focus on visualizing stock prices rather than analysing how trading strategies perform during bullish, bearish, or sideways markets. As a result, there is a need for a more intelligent and data-driven system that can evaluate strategies, measure risks, and provide meaningful investment insights.

To address these challenges, this project proposes an Indian Market Portfolio Intelligence and Back testing Platform that combines financial analytics with machine learning techniques.. It supports multiple trading strategies such as Buy and Hold, Moving Average Crossover, RSI-based trading, Momentum Strategy, and Mean Reversion Strategy.

The system retrieves historical stock market data from Yahoo Finance and processes it to generate important financial features such as returns, volatility, momentum, and drawdown. Machine learning algorithms are integrated into the platform to detect market regimes and recommend suitable strategies based on prevailing market conditions. The project also performs portfolio analysis and risk assessment using important performance metrics like Sharpe Ratio, cumulative returns, and volatility measures.

The proposed platform serves as a valuable tool for investors, financial researchers, and students interested in stock market analytics. By integrating strategy evaluation, machine learning, and risk analysis into a single system, the project aims to improve investment decision-making and provide a deeper understanding of financial market behavior in the Indian stock market environment.

Chapter 2

LITERATURE SURVEY

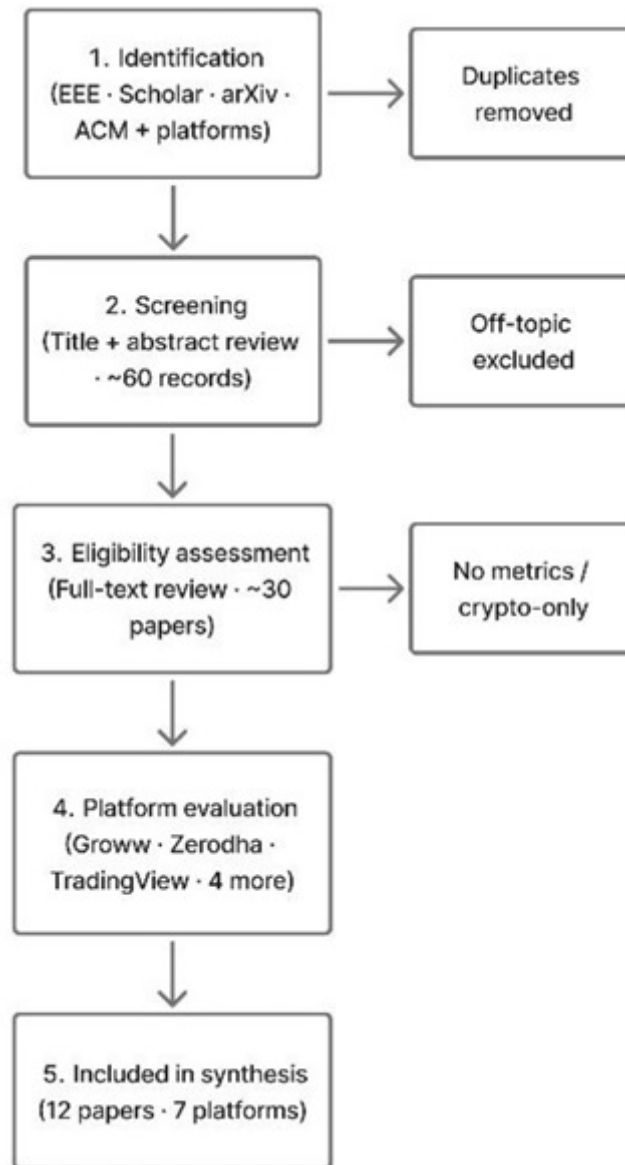


Fig 2.1: Survey Methodology Pipeline

2.1 - Enhancing Stock Market Prediction: A Robust LSTM-DNN Model Analysis on 26 Real-Life Datasets

Authors: K. Alam, M. H. Bhuiyan, I. U. Haque, M. F. Monir, and T. Ahmed

Journal: IEEE Access

Volume & Issue: Vol. 12, 2024

Summary:

This paper presents a hybrid LSTM-DNN framework for stock market prediction using multiple real-world financial datasets. The system combines Long Short-Term Memory (LSTM) networks and Deep Neural Networks (DNN) to improve stock forecasting accuracy and handle nonlinear market behavior effectively. The study demonstrated that hybrid deep learning models outperform traditional forecasting methods in analyzing dynamic financial time-series data.

Methodology:

The authors trained a hybrid LSTM-DNN architecture using historical stock market datasets and technical indicators. LSTM layers captured temporal dependencies while DNN layers extracted nonlinear patterns from market data. The model was evaluated across 26 real-life datasets to test robustness and prediction capability.

Outcomes:

- Improved stock price prediction accuracy compared to traditional methods.
- Demonstrated effective handling of nonlinear and sequential market data.
- Showed strong forecasting performance across multiple datasets.
- Limited mainly to prediction without integrated portfolio analytics or risk-aware strategy evaluation.

2.2 Stock Price Prediction Using CNN and LSTM-Based Deep Learning Models

Authors: S. Mehtab and J. Sen

Conference: International Conference on Decision Aid Sciences and Application (DASA)

Year: 2020

Summary:

This paper proposed a CNN-LSTM-based deep learning system for stock market prediction. The study combined Convolutional Neural Networks (CNN) for extracting local market features and LSTM networks for learning long-term temporal dependencies in stock market data. The model demonstrated improved prediction capability compared to standalone machine learning approaches.

Methodology:

The framework used historical stock market datasets and technical indicators as input data. CNN layers extracted local trend patterns from stock price movements, while LSTM layers

learned sequential dependencies over time. The combined model was trained and tested on financial time-series data for stock forecasting tasks.

Outcomes:

- Improved prediction performance using hybrid CNN-LSTM architecture.
- Captured both local and temporal market patterns effectively.
- Demonstrated strong capability in analyzing financial time-series data.
- Performance remained sensitive to sudden market volatility and regime changes.

2.3 An Application of Deep Reinforcement Learning to Algorithmic Trading

Authors: T. Théate and D. Ernst

Journal: Expert Systems with Applications

Volume & Issue: Vol. 173, 2021

Summary:

This paper explored the application of Deep Reinforcement Learning (DRL) in algorithmic trading systems. The proposed framework enabled intelligent trading agents to learn adaptive trading strategies directly from market environments through reward optimization techniques. The study demonstrated that DRL systems can outperform static rule-based trading systems under simulated market conditions.

Methodology:

The authors utilized Deep Q-Networks (DQN) and policy optimization methods using OHLCV market data and technical indicators such as RSI and MACD. Trading agents interacted with simulated market environments and continuously updated their strategies based on portfolio rewards and trading performance.

Outcomes:

- Enabled adaptive automated trading strategy generation.
- Improved trading decision-making under varying market conditions.
- Demonstrated better adaptability compared to rule-based systems.
- Faced limitations such as overfitting, computational complexity, and transaction cost sensitivity.

2.4 A Hybrid Learning Approach to Detecting Regime Switches in Financial Markets

Authors: P. Akioyamen, Y. Z. Tang, and H. Hussien

Conference: ACM International Conference on AI in Finance (ICAIF '20)

Year: 2020

Summary:

This paper proposed a hybrid machine learning approach for detecting regime switches in financial markets. The research focused on identifying bullish, bearish, and sideways market conditions using unsupervised learning and statistical analysis. The study highlighted that regime-aware systems improve trading strategy robustness and portfolio adaptability.

Methodology:

The authors applied clustering techniques on financial market datasets containing volatility, returns, and trend indicators. Machine learning models identified hidden structural patterns in market behavior to classify different market regimes. The system combined statistical analysis and unsupervised learning for improved regime detection under non-stationary conditions.

Outcomes:

- Successfully detected financial market regime switches.
- Improved contextual trading strategy evaluation and portfolio analysis.
- Enhanced adaptability under changing market conditions.
- Existing systems lacked integration with unified portfolio analytics platforms.

2.5 Backtest Overfitting in the Machine Learning Era: A Comparison of Out-of-Sample Testing Methods in a Synthetic Controlled Environment

Authors: H. Arian, D. Norouzi Mobarekeh, and L. Seco

Journal: Knowledge-Based Systems

Volume & Issue: Vol. 305, 2024

Summary:

This paper investigated the problem of backtest overfitting in financial machine learning systems. The study emphasized that many trading strategies perform well on historical data but fail in real-world market environments due to poor validation techniques and data leakage. The research highlighted the importance of robust out-of-sample testing for reliable financial model evaluation.

Methodology:

The authors compared multiple out-of-sample validation techniques using synthetic financial environments and machine learning-based trading systems. The framework analyzed how

improper validation methods affect strategy reliability and model generalization under non-stationary market conditions.

Outcomes:

- Identified severe risks of backtest overfitting in financial ML systems.
- Demonstrated the importance of proper validation frameworks.
- Improved understanding of out-of-sample testing reliability.
- Highlighted the need for risk-aware and regime-aware evaluation systems for financial analytics platforms.

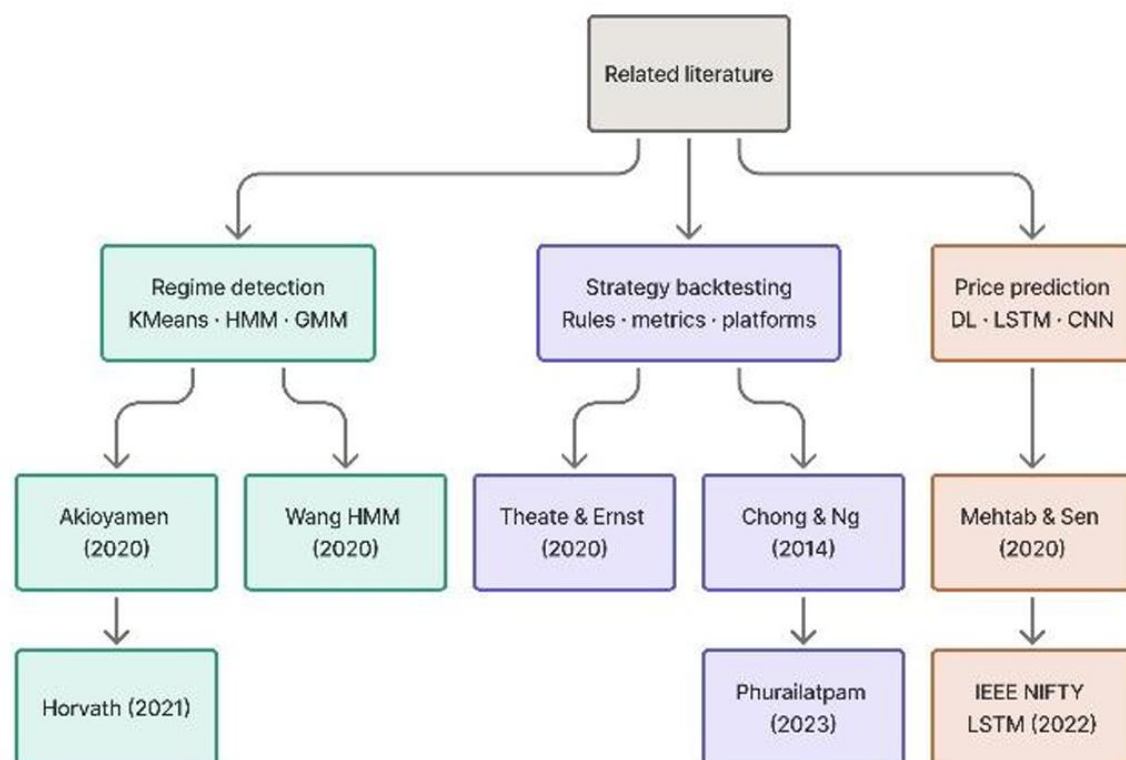


Fig 2.2: Classification of surveyed literature in intelligent financial systems.

Chapter 3

PROBLEM IDENTIFICATION

Financial markets are highly volatile, making investment decisions difficult for investors. Many traders choose strategies without properly analysing risk, market conditions, or historical performance, which can lead to financial losses. Existing stock market platforms mainly focus on price charts and basic analysis but do not provide proper strategy evaluation or risk assessment.

Another major issue is the lack of intelligent systems that can analyse market conditions and recommend suitable trading strategies. Investors also find it difficult to measure important factors such as volatility, drawdown, and overall portfolio performance using traditional methods.

Therefore, there is a need for a smart and data-driven platform that can perform strategy back testing, market analysis, and risk evaluation using historical stock market data and machine learning techniques. The proposed Indian Market Portfolio Intelligence and Back testing Platform aims to address these challenges by helping users analyse and evaluate trading strategies more effectively

Chapter 4

GOALS AND OBJECTIVES

4.1 GOALS

1. To develop a machine learning–based platform for analysing and evaluating trading strategies using historical stock market data.
2. To provide accurate back testing of different trading strategies under various market conditions.
3. To detect market regimes such as bullish, bearish, and sideways markets using data analysis techniques.
4. To help investors understand portfolio performance, risk, and return through financial metrics.
5. To provide intelligent strategy recommendations for better investment decision-making.
6. To create a useful learning and research tool for financial analytics and stock market analysis.

4.2 OBJECTIVES

- 1. To develop a strategy back testing system**

To create a reliable back testing engine that evaluates predefined trading strategies using historical stock market data and measures their effectiveness over different time periods.

- 2. To collect and process historical market data**

To retrieve stock market data from financial sources such as Yahoo Finance and preprocess it for analysis by handling missing values, feature extraction, and data formatting.

- 3. To implement multiple trading strategies**

To integrate various trading strategies such as Buy and Hold, Moving Average Crossover, RSI Strategy, Momentum Strategy, and Mean Reversion Strategy for comparative analysis.

- 4. To detect market regimes using machine learning**

To apply unsupervised machine learning techniques for identifying different

market conditions such as bullish, bearish, and sideways markets based on historical trends and volatility.

5. To evaluate strategy performance under different market conditions

To analyse how each trading strategy performs in varying market environments and determine its strengths and weaknesses.

Chapter 5

SYSTEM REQUIREMENT SPECIFICATION

5.1 Hardware Requirements

- **Processor:** Intel Core i5/i7 or AMD Ryzen processor
- **RAM:** 16 GB or higher for faster data processing and machine learning tasks
- **Storage:** SSD with 10 GB or more free space
- **Display:** Full HD monitor for better visualization of charts and analytics

5.2 Software Requirements

- **Operating System**-Windows 10/11, Linux distributions (Ubuntu or Fedora)
macOS
- **Programming Language**-Python 3.10 or above
- **Development Tools**-Visual Studio Code (VS Code) / PyCharm, Jupiter Notebook
for data analysis and testing
- **Libraries and Frameworks**
 - Pandas -Data manipulation and analysis
 - NumPy - Numerical computations
 - Scikit-learn -Machine learning algorithms
 - SciPy -Scientific and statistical analysis
 - Plot / Matplotlib -Data visualization and chart generation
 - yFinance / NSEpy -Stock market data retrieval
 - FastAPI -Backend API development
- **Database**-PostgreSQL for storing user data, strategy results, and portfolio
information
- **Web Technologies**-HTML, CSS, JavaScript for frontend development

Chapter 6

METHODOLOGY

The proposed Indian Market Portfolio Intelligence and Backtesting System is a regime-aware AI-driven financial analytics platform designed to evaluate trading strategies under different market conditions using machine learning and portfolio analysis techniques. The system integrates market regime detection, historical backtesting, strategy evaluation, and risk forecasting into a unified framework for Indian equity markets

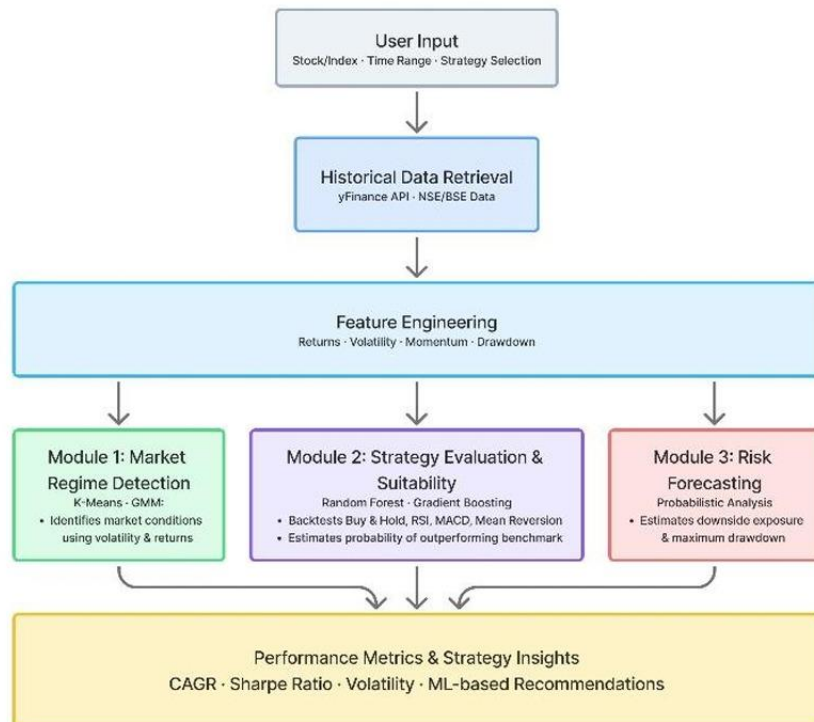


Fig: 6.1 Architecture Diagram

6.1 Data Collection

Historical stock market data is collected using the Yahoo Finance API through the yFinance library. The system retrieves OHLCV (Open, High, Low, Close, Volume) data for selected stocks and indices based on the user's preferred time range.

6.2 Data Processing

Raw market data may contain missing values, duplicate records, or inconsistencies. Therefore, preprocessing is performed to improve the quality and reliability of the dataset.

The preprocessing stage includes:

- Handling missing or null values
- Removing duplicate records

- Formatting date and time values
- Normalizing financial data
- Organizing the dataset into structured tabular form

This step ensures that the data becomes suitable for feature extraction, machine learning, and strategy analysis.

6.3 Feature Engineering

Feature engineering is one of the most important stages in the system. In this stage, meaningful financial indicators are generated from historical stock data.

The platform calculates features such as:

- Daily Returns
- Volatility
- Momentum
- Moving Averages
- Relative Strength Index (RSI)
- Maximum Drawdown
- Rolling Mean and Standard Deviation

These features help the system identify stock market trends, price movements, and investment risk. The generated features are later used for market regime detection, machine learning models, and strategy recommendations.

6.4 Market Regime Detection

Financial markets behave differently under various conditions. Therefore, the system identifies market regimes such as:

- Bullish Market
- Bearish Market
- Sideways Market

Unsupervised machine learning algorithms like clustering techniques are used to analyze hidden patterns in market behavior. The model studies features such as returns, volatility, and momentum to classify the current market condition.

This helps investors understand the nature of the market before applying any trading strategy.

6.5. Trading Strategy Implementation

The platform implements multiple trading strategies that are commonly used in stock market analysis. Each strategy follows a different trading logic and investment approach.

The strategies include:

- **Buy and Hold Strategy**
 - Investors purchase stocks and hold them for a long period regardless of short-term market fluctuations.
- **Moving Average Crossover Strategy**
 - Buy and sell signals are generated when short-term moving averages cross long-term moving averages.
- **RSI Strategy**
 - The Relative Strength Index is used to identify overbought and oversold conditions in stocks.
- **Momentum Strategy**
 - Stocks with strong recent performance are selected with the expectation that the trend will continue.
- **Mean Reversion Strategy**
 - This strategy assumes that stock prices eventually return to their historical average values.
- Each strategy is tested independently using historical market data.

6.6 Backtesting Process

Backtesting is used to evaluate how a trading strategy would have performed in past market conditions. The platform simulates real-time trading using historical stock data.

During backtesting, the system:

- Generates buy and sell signals
- Executes virtual trades
- Tracks profits and losses
- Calculates portfolio growth over time

The backtesting engine helps determine whether a strategy is profitable, stable, and suitable for different market environments.

6.7 Performance and Risk Analysis

After executing the trading strategies, the platform performs detailed performance and risk analysis.

The following financial metrics are calculated:

- Cumulative Returns
- Annual Returns
- Volatility
- Sharpe Ratio
- Maximum Drawdown
- Risk-Adjusted Returns

These metrics help users understand:

- Profitability of strategies
- Risk exposure
- Portfolio stability
- Long-term investment performance

Risk analysis is important because a strategy with high returns may also involve high financial risk.

6.8 Machine Learning–Based Strategy Recommendation

The platform uses supervised machine learning models to recommend suitable trading strategies based on historical performance and current market conditions.

The model is trained using:

- Historical financial indicators
- Market regimes
- Strategy performance metrics

Based on the analysed patterns, the system predicts which strategy may perform better under current market conditions.

This feature provides intelligent decision support to investors and improves investment planning.

6.9 Result Visualization and Reporting

The final stage of the methodology involves presenting results in a clear and user-friendly format.

The platform provides:

- Performance graphs
- Portfolio growth charts
- Risk comparison charts
- Strategy comparison reports
- Market trend visualizations

Visualization helps users easily understand:

- Market behavior
- Strategy performance
- Portfolio risk
- Investment outcomes

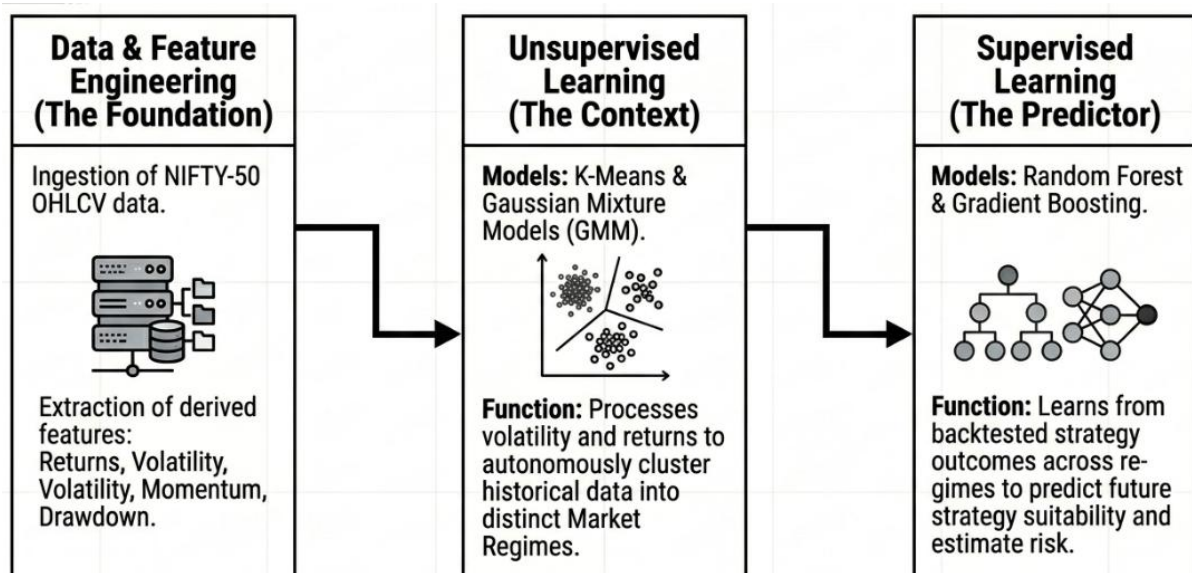


Fig: 6.2 Machine Learning Pipeline

Chapter 7

APPLICATIONS

1. Investment Decision Support

The platform helps investors analyze and compare different trading strategies before investing in the stock market. By providing data-driven insights and performance metrics, users can make better financial decisions with reduced risk.

2. Portfolio Management

The system assists users in managing and evaluating stock portfolios by analysing returns, volatility, and overall portfolio performance. It helps investors understand how different assets contribute to profit and risk.

3. Strategy Backtesting

Traders and analysts can test trading strategies on historical market data to evaluate their profitability and effectiveness before applying them in real market conditions.

4. Risk Analysis and Management

The platform calculates important financial metrics such as Sharpe Ratio, volatility, and maximum drawdown. This helps users identify potential risks and choose safer investment strategies.

5. Market Regime Analysis

Machine learning techniques are used to identify bullish, bearish, and sideways market conditions. This helps investors select suitable strategies based on current market behavior.

6. Educational and Learning Tool

The project serves as an educational platform for students and beginners who want to learn stock market analytics, trading strategies, portfolio management, and machine learning applications in finance.

Chapter 8

CONTRIBUTION TO SOCIETY AND ENVIRONMENT

1.Improves Financial Awareness

The platform helps users understand stock market behavior, investment strategies, and financial risk management, thereby increasing financial literacy among students and investors.

2.Supports Better Investment Decisions

By providing data-driven insights and strategy recommendations, the system helps investors make informed financial decisions and reduce unnecessary financial losses.

3.Educational Benefits

The project acts as a learning tool for students, researchers, and beginners interested in financial analytics, machine learning, and portfolio management.

4.Encourages Technology In Finance

The platform demonstrates the use of machine learning and data analysis in the financial sector, promoting innovation and technological advancement.

5.Assists Small Investors

Retail investors with limited market knowledge can use the platform to evaluate trading strategies and understand market risks more effectively.

6.Reduces Paper Usage

The platform is completely digital, reducing the need for printed reports, financial records, and manual documentation

7.Efficient Data Management

Digital storage and automated processing help reduce unnecessary manual operations and resource wastage

8.Minimizes Physical Infrastructure Requirements

The platform reduces dependency on traditional paperwork and offline analysis methods, contributing to an eco-friendlier working environment

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APPENDIX-I

PUBLISHED SURVEY PAPER



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DOI: 10.55041/ISJEM07359

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A Survey on Regime-Aware Portfolio Analytics and ML-Based Strategy Evaluation for Indian Equity Markets

Mrs. Beena K¹, Jayaditya Dev², Durgashree M³, Aryaman Tiwari³, Chirag T⁵

Assistant Professor, Dept of CSE, KSIT, Karnataka, India¹

Student, Dept of CSE, KSIT, Karnataka, India²⁻⁵

ABSTRACT – The increasing complexity and volatility of financial markets have created a growing demand for intelligent systems capable of accurate stock prediction, adaptive trading, and efficient risk management. Traditional statistical methods and many existing financial platforms primarily focus on visualization and isolated backtesting, often lacking integrated regime-aware and risk-aware analytical capabilities. This survey reviews recent advancements in machine learning, deep learning, and reinforcement learning techniques applied to stock market prediction, algorithmic trading, market regime detection, portfolio optimization, and financial stress testing. The study examines widely used models such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Autoencoders, and Deep Reinforcement Learning (DRL). Additionally, emerging approaches involving sentiment analysis, cross-market data integration, and regime clustering are discussed. Key challenges including backtest overfitting, data non-stationarity, and model generalization are also analyzed. The reviewed literature indicates that hybrid machine learning approaches can significantly improve forecasting accuracy and trading performance compared to traditional techniques, while challenges related to robustness and real-time adaptability remain important research areas.

KEY WORDS: Stock Market Prediction, Algorithmic Trading, Machine Learning, Deep Learning, Reinforcement Learning, Market Regime Detection, Financial Risk Management, Sentiment Analysis, Portfolio Optimization

1. INTRODUCTION

Financial markets are highly dynamic, nonlinear, and influenced by numerous economic, political, and psychological factors. Predicting stock price movements and evaluating investment strategies have therefore become major challenges in quantitative finance and financial analytics. Traditional statistical and econometric methods often struggle to model the complex temporal dependencies, volatility fluctuations, and regime shifts present in financial time-series data. As a result, machine learning and deep learning approaches have gained significant attention due to their ability to identify hidden patterns and learn from large volumes of historical market data. Recent studies have demonstrated the effectiveness of models such as Long Short-Term Memory (LSTM),

Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Reinforcement Learning (RL) in improving stock market forecasting and automated trading performance.

The rapid growth of algorithmic trading and AI-driven financial systems has also increased the importance of portfolio intelligence, market regime detection, and risk-aware strategy evaluation. Existing financial platforms primarily focus on chart visualization and isolated backtesting functionalities, often lacking integrated systems capable of combining market prediction, regime analysis, strategy recommendation, and downside risk forecasting within a unified framework. Furthermore, financial datasets are characterized by non-stationarity, autocorrelation, noise, and sudden regime transitions, which make reliable model evaluation particularly difficult.

Several recent studies have explored advanced approaches for addressing these challenges. Hybrid deep learning architectures combining LSTM, DNN, and sentiment analysis have shown improved predictive capability for stock market forecasting. Reinforcement learning techniques have also been applied to algorithmic trading for adaptive decision-making under uncertain market conditions. In parallel, unsupervised learning methods such as Hidden Markov Models (HMM), clustering techniques, and Wasserstein-distance-based regime detection have emerged as effective tools for identifying changing market conditions and improving trading strategy robustness.

Another major concern in financial machine learning is backtest overfitting, where models perform well on historical data but fail to generalize in real market environments. Recent research emphasizes the need for specialized validation frameworks, robust cross-validation techniques, and probabilistic risk analysis for reliable financial model evaluation.

This survey reviews recent advancements in machine learning-driven financial market systems with a focus on stock market prediction, algorithmic trading, market regime detection, portfolio intelligence, and risk forecasting. The study analyzes major methodologies, compares their strengths and limitations, and identifies current research gaps in developing adaptive and robust financial analytics platforms.

2. METHODS



The methodological framework adopted in this survey follows a systematic literature review approach aimed at gathering, analyzing, and synthesizing existing research on stock market prediction, portfolio analytics, market regime detection, and strategy evaluation systems. The primary research question guiding this survey is: What methods and platforms currently exist for evaluating financial trading strategies under varying market conditions, and what gaps remain, particularly for Indian equity markets? The survey further investigates the limitations of existing financial platforms in providing integrated regime-aware and risk-aware portfolio intelligence.

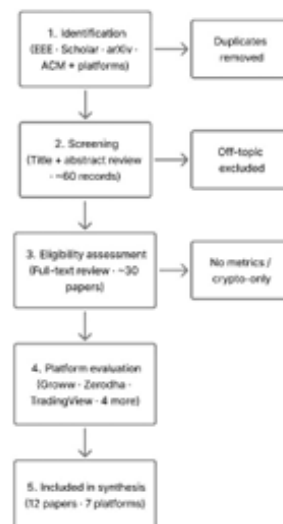
2.1 Search Strategy - Literature search was conducted across IEEE Xplore, Google Scholar, arXiv, ScienceDirect, and ACM Digital Library using keyword combinations including "stock market prediction using LSTM," "algorithmic trading machine learning," "market regime detection," "portfolio backtesting framework," "risk-adjusted strategy evaluation," and "Indian equity market analytics." Commercial platforms such as Zerodha, Streak, Groww, TradingView, Smallcase, INDmoney, and QuantConnect were additionally evaluated through official documentation and comparative feature analysis. Research published between 2018 and 2025 was prioritized, along with selected foundational studies relevant to financial machine learning and regime analysis.

2.2 Inclusion and exclusion criteria - The survey included papers related to stock market prediction, algorithmic trading, market regime detection, portfolio evaluation, financial stress testing, and machine learning-based risk analysis. Studies involving LSTM, CNN, reinforcement learning, clustering algorithms, Hidden Markov Models, and hybrid deep learning architectures were prioritized due to their relevance to intelligent financial systems. Papers focused solely on theoretical finance without computational implementation, cryptocurrency-only systems lacking broader equity applicability, duplicate studies, and publications without quantitative evaluation metrics were excluded. Platform analysis was limited to analytical and backtesting capabilities rather than brokerage-specific operations.

2.3 Analysis Framework - The selected studies were evaluated based on methodology, datasets used, prediction accuracy, validation techniques, risk metrics, computational complexity, and adaptability to changing market conditions. Additional emphasis was placed on integration between prediction models, regime detection systems, portfolio backtesting, and strategy recommendation frameworks. Existing financial platforms were comparatively analyzed based on support for strategy backtesting, portfolio-level evaluation, regime detection, risk-adjusted performance metrics, machine learning integration, Indian market compatibility, and accessibility for non-technical users. The final synthesis enabled identification of current technological gaps and established the foundation for a

proposed integrated regime-aware portfolio intelligence framework.

Fig-1: Survey Methodology Flowchart



3. RESULTS

The literature review identified five major research domains within intelligent financial market systems: stock market prediction, algorithmic trading, market regime detection, portfolio backtesting and risk evaluation, and existing financial analytics platforms.



Fig-2: Classification of surveyed literature in intelligent financial systems.

The reviewed studies demonstrate that modern machine learning approaches significantly improve predictive capability and adaptive decision-making compared to traditional statistical methods. However, major limitations still exist in terms of integration, robustness, interpretability, and applicability to Indian equity markets.



3.1 Stock Market Prediction Models - A large portion of recent financial research focuses on predicting stock price movements using machine learning and deep learning techniques. Mehtab and Sen (2020) proposed an LSTM-DNN hybrid architecture for stock market prediction across multiple real-world datasets, demonstrating improved prediction accuracy compared to traditional forecasting methods. Similarly, Phurailatpam et al. (2022) utilized CNN-LSTM-based architectures for stock price prediction using historical financial indicators and time-series data.

Several studies also incorporated multi-market and cross-domain information into predictive frameworks. Cabrera et al. (2024) proposed a multi-market data-driven prediction framework that combined optimization techniques with deep learning architectures for enhanced forecasting performance across multiple stock exchanges. In another study, Wang et al. (2024) integrated social media sentiment analysis with Transductive Long Short-Term Memory (TLSTM) networks, demonstrating improved prediction capability under volatile market conditions.

Traditional technical indicators continue to remain relevant in financial forecasting. Chong and Ng (2008) revisited the performance of MACD and RSI oscillators, showing that momentum-based indicators still retain predictive significance under certain market conditions despite the rise of deep learning models.

Although these approaches achieved improved predictive accuracy, most systems remained limited to isolated forecasting tasks without integration into portfolio-level strategy evaluation or regime-aware investment analysis.

3.2 Algorithmic Trading and Reinforcement Learning - Recent advancements in reinforcement learning have enabled adaptive trading systems capable of learning optimal trading strategies directly from market environments. Moody and Saffell (2001) pioneered reinforcement learning applications in trading through direct reinforcement frameworks, while more recent work by Théate and Ernst (2020) applied Deep Reinforcement Learning (DRL) to algorithmic trading using Deep Q-Networks and policy optimization approaches.

Algorithmic trading systems commonly utilize OHLCV data, technical indicators such as RSI and MACD, and portfolio state variables for decision-making. Several studies demonstrated that reinforcement learning agents can outperform static rule-based strategies under simulated market environments by adapting to changing price dynamics.

However, the reviewed literature also highlighted major practical limitations. Many reinforcement learning systems rely heavily on simplified reward structures, simulated trading environments, and computationally expensive training pipelines. Real-world deployment challenges such as transaction costs, slippage, overfitting, and changing market regimes remain insufficiently addressed.

3.3 Market Regime Detection Techniques - Market regime detection has emerged as an important research area for improving investment strategy evaluation and portfolio adaptability. Hidden Markov Models (HMM), K-Means clustering, Gaussian Mixture Models (GMM), and Wasserstein-distance-based clustering techniques have been widely explored for identifying bull, bear, and sideways market regimes.

Akiyamen et al. (2020) proposed unsupervised clustering approaches for identifying financial market states based on volatility and return characteristics. Wang et al. (2020) further demonstrated the effectiveness of Hidden Markov Models in regime-switching financial systems for adaptive portfolio allocation. Horvath et al. (2021) utilized Wasserstein distance-based clustering for improved separation of financial market regimes under non-stationary conditions.

Additionally, regime-aware factor investing frameworks using Hidden Markov Models were explored by several studies for tactical portfolio management and adaptive asset allocation. Hybrid learning approaches combining supervised and unsupervised methods also demonstrated improved adaptability for detecting financial regime switches.

Despite promising results, most existing regime detection systems operate independently from portfolio analytics and trading strategy recommendation frameworks, limiting their practical applicability for retail investors and non-technical users.

3.4 Portfolio Backtesting and Risk Evaluation - Robust portfolio backtesting and risk-adjusted evaluation remain critical components of financial machine learning systems. Several studies emphasized the importance of performance metrics such as Sharpe Ratio, Sortino Ratio, CAGR, volatility, and maximum drawdown for evaluating trading strategies under varying market conditions.

Lopez de Prado (2018) highlighted the growing problem of backtest overfitting in machine learning-based financial systems and introduced advanced validation techniques such as Purged K-Fold Cross-Validation and Combinatorial Purged Cross-Validation to improve out-of-sample reliability.

In parallel, stress-testing frameworks utilizing machine learning approaches were proposed for evaluating Indian financial market portfolios under adverse economic conditions. Regime-aware portfolio optimization methods also demonstrated improved robustness by dynamically adapting trading strategies according to changing volatility and macroeconomic environments.

However, many existing systems provide only isolated backtesting functionality without integrating regime detection, probabilistic risk forecasting, or machine learning-driven strategy suitability analysis.

3.5 Comparative Analysis and Identified Gaps - Table 1 summarizes the major approaches identified across the surveyed literature, including stock market prediction, algorithmic trading, market regime detection,



and portfolio evaluation systems. The reviewed studies demonstrate that deep learning architectures such as LSTM, CNN-LSTM, and hybrid DNN frameworks achieve improved forecasting accuracy compared to traditional statistical approaches. Reinforcement learning systems further improve adaptability by dynamically adjusting trading strategies under changing market conditions. Similarly, Hidden Markov Models, clustering algorithms, and Wasserstein-distance-based approaches have shown effectiveness in identifying market regimes and improving tactical portfolio allocation.

Despite these advancements, the literature reveals several important limitations. Most stock prediction systems focus only on forecasting accuracy without integrating strategy evaluation or portfolio-level analytics. Reinforcement learning frameworks often rely on simulated environments and computationally expensive training procedures, reducing their practical applicability in real-world trading. Regime detection models generally operate independently from backtesting systems and rarely provide interpretable investment insights for non-technical users. Additionally, existing financial platforms such as TradingView, Zerodha/Streak, and QuantConnect provide isolated functionalities such as chart visualization, manual backtesting, or algorithmic execution, but lack integrated regime-aware portfolio intelligence specifically designed for Indian equity markets.

Another major concern identified in the reviewed literature is the problem of backtest overfitting and poor out-of-sample generalization. Lopez de Prado (2018) demonstrated that many machine learning-based financial systems fail under non-stationary market conditions due to improper validation techniques and data leakage. Similarly, stress-testing frameworks for Indian financial portfolios remain limited in accessibility and integration with retail trading platforms.

The biggest gap identified across all surveyed domains is the absence of a unified framework capable of combining stock prediction, market regime detection, portfolio backtesting, machine learning-based strategy evaluation, and risk-aware analytics within a single accessible platform. Most existing systems specialize in only one domain and do not provide integrated decision support for adaptive portfolio management under varying market conditions. This gap becomes even more significant in the context of Indian equity markets, where regime-aware retail analytics platforms remain limited.

Table 1: Comparative Summary of Surveyed Approaches

Approach	Method	Key Strength	Limitation
Stock Prediction	LSTM, CNN, DNN	Accurate forecasting	Limited integration
Algorithmic Trading	RL, Indicator Strategies	Adaptive trading	Overfitting risk
Regime Detection	HMM, K-Means, GMM	Detects market states	Standalone usage

Backtesting & Risk	Sharpe, Drawdown, CV	Strategy evaluation	Validation issues
Existing Platforms	TradingView, Zerodha	Visualization support	No regime analytics
Proposed Framework	ML + Regime + Backtest	Unified analysis	Needs real-world testing

3.6 Proposed Integrated System Overview - Based on the identified gaps, an integrated regime-aware portfolio intelligence framework is proposed that combines historical market data retrieval, market regime detection, portfolio backtesting, strategy evaluation, and risk analysis within a unified system. The framework utilizes Yahoo Finance data pipelines, technical indicators, and predefined strategies such as Buy-and-Hold, RSI-based trading, and Moving Average Crossover strategies. K-Means clustering is used to identify bull, bear, and sideways market conditions. The architecture combines a FastAPI backend with a Next.js frontend to provide interactive portfolio analytics, equity curve visualization, regime analysis, and ML-based strategy recommendations for Indian equity markets.

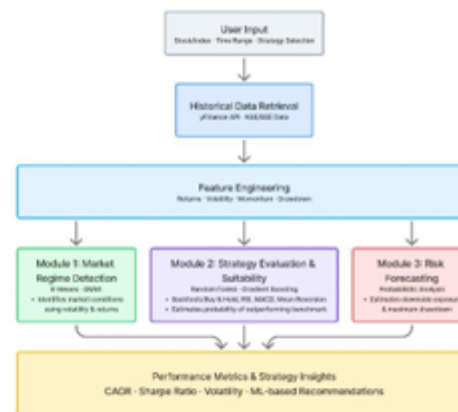


Fig. 2: Proposed Integrated System Architecture

Table 2: Expected System Performance Against Baseline

Metric	Baseline (Existing Platforms)	With Proposed System
Strategy Evaluation	Manual / isolated testing	Regime-aware evaluation
Market Analysis	Static indicators	Adaptive ML-based insights
Portfolio Backtesting	Limited strategy scope	Multi-strategy backtesting
Risk Assessment	Basic metrics only	Integrated risk analytics
Regime Detection	Not supported	Bull/Bear/Sideways detection



User Accessibility	Technical or fragmented	Unified interactive dashboard
Indian Market Support	Partial support	Indian equity focused
Strategy Recommendation	User-driven decisions	ML-assisted recommendations

4. DISCUSSION

The reviewed literature demonstrates that machine learning and deep learning techniques have significantly improved financial market prediction, trading strategy evaluation, and market regime analysis. However, research in this domain remains highly fragmented, with most studies focusing on isolated components such as stock forecasting, reinforcement learning-based trading, regime detection, or portfolio backtesting independently. Although each area has produced technically effective solutions, very few systems integrate these capabilities into a unified framework suitable for practical investment decision-making, particularly within Indian equity markets.

One major observation from the survey is that prediction accuracy alone is insufficient for building effective financial analytics systems. Several studies report strong forecasting performance using LSTM, CNN-LSTM, and reinforcement learning models, yet many of these approaches lack interpretability, regime awareness, and portfolio-level applicability under real market conditions. Additionally, financial markets are inherently non-stationary, making models highly sensitive to changing volatility patterns and economic conditions. As highlighted by Lopez de Prado (2018), improper validation methodologies can lead to severe backtest overfitting and unreliable out-of-sample performance.

Another important issue identified in the literature is the limited availability of integrated financial datasets. Real-world deployment of intelligent portfolio systems requires synchronized access to historical price data, macroeconomic indicators, technical signals, volatility measures, and potentially sentiment-based information sources. However, many existing studies rely on simulated datasets, restricted market samples, or isolated benchmark environments, reducing practical generalizability. Similarly, commercial platforms provide strong visualization and charting support but rarely incorporate regime-aware machine learning pipelines or adaptive strategy recommendation systems.

The proposed framework addresses existing limitations by integrating market regime detection, portfolio backtesting, strategy evaluation, and risk-aware analytics within a unified architecture. K-Means clustering is used for regime identification due to its computational simplicity and interpretability, although future work may explore advanced probabilistic approaches such as Hidden Markov Models and Gaussian

Mixture Models for improved regime consistency and real-time adaptation. Unlike existing platforms focused primarily on isolated visualization or manual backtesting, the proposed system combines predictive analytics with portfolio-level strategy evaluation and regime-aware decision support within a single accessible platform tailored for Indian equity markets.

5. CONCLUSION

The literature survey examined major research areas within intelligent financial market systems, including stock market prediction, algorithmic trading, market regime detection, portfolio backtesting, and risk-aware portfolio evaluation. The reviewed studies demonstrate that machine learning and deep learning approaches such as LSTM, reinforcement learning, clustering algorithms, and Hidden Markov Models significantly improve forecasting accuracy and adaptive trading performance compared to traditional statistical methods. However, most existing studies and commercial platforms remain focused on isolated functionalities such as prediction, visualization, or strategy testing without providing integrated regime-aware portfolio intelligence.

The survey identified major challenges including backtest overfitting, poor out-of-sample generalization, fragmented analytics pipelines, and limited accessibility of advanced quantitative tools for retail investors. Existing platforms such as TradingView, Zerodha/Streak, and QuantConnect provide strong charting and backtesting capabilities but lack integrated machine learning-driven systems tailored for Indian equity markets.

Based on the identified gaps, this paper proposed a regime-aware portfolio intelligence framework integrating market regime detection, portfolio backtesting, machine learning-based strategy evaluation, and risk analytics into a single accessible architecture. The proposed framework demonstrates the feasibility of combining predictive analytics with adaptive strategy evaluation for Indian financial markets.

Future work may focus on incorporating real-time streaming market data, probabilistic regime-switching models, sentiment-aware forecasting, and advanced reinforcement learning strategies. Large-scale deployment and real-world validation across broader market conditions will also be necessary to evaluate scalability, robustness, and practical effectiveness. Overall, this survey establishes a structured understanding of current intelligent financial analytics systems and highlights the need for integrated regime-aware decision-support platforms for modern financial markets.

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APPENDIX-II

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APPENDIX - III

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